

Prompt Engineering for Image Captioning in Remote Sensing using PaliGemma

DI725 Term Project Phase 1 Report

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Abstract—This project explores prompt engineering and inference benchmarking for satellite image captioning using Google’s PaliGemma model. Originally designed as a fine-tuning study, the project shifted to an inference-only approach due to hardware constraints. Through three stages of prompt refinement and an ablation involving decoding parameters and prompt length cues, I analyze refusal rates and output quality. Results show that prompt phrasing significantly affects caption success in zero-shot settings, revealing both opportunities and limitations of frozen vision-language models.

I. INTRODUCTION

Image captioning is a key vision-language task that translates visual information into descriptive natural language. With the release of large pre-trained vision-language models (VLMs) like PaliGemma, opportunities arise to build captioning systems with minimal or no training. However, inference quality can vary significantly depending on prompt formulation and decoding parameters.

II. LITERATURE REVIEW

Text augmentation has proven to be an effective tool in various natural language processing tasks. I reviewed the following relevant works that inform my approach:

A. Back-Translation for Data Augmentation

Sennrich et al. (2016) demonstrated that back-translation could be used to synthesize high-quality data for machine translation tasks. This technique has since been generalized to augment datasets for various NLP tasks. In image captioning, it helps by generating natural sentence variants that preserve semantic meaning, which helps diversify the dataset and improve training quality.

B. Paraphrasing with Pretrained Transformers

Raffel et al. (2020) introduced the T5 model, a versatile text-to-text transformer that includes paraphrasing capabilities. Similarly, Pegasus (Zhang et al., 2020) pre-trains on sentence gaps for summarization and paraphrase generation. These models enable the generation of caption variants, promoting greater linguistic richness in training.

C. Image Captioning Metrics

Zhao et al. (2021) detailed the metrics commonly used to evaluate image captioning tasks such as BLEU, METEOR, ROUGE-L, CIDEr, and SPICE. These methods proved to be more reliable than using human ratings, being less subjective and easier to use on a large scale.

D. Opportunities in Captioning Models

While architectural improvements like LoRA (Hu et al., 2022) allow efficient transformer fine-tuning, simple preprocessing-based enhancements remain underexplored in the context of vision-language models. Text augmentation provides a promising, low-cost opportunity to improve training without increasing computational complexity.

These papers indicate that while complex fine-tuning methods dominate the research, there is a research gap in leveraging simple augmentation strategies for improving captioning diversity, especially in the context of remote sensing.

III. PROJECT PROPOSAL

A. Original Research Question

How can simple text augmentation methods like paraphrasing and back-translation improve the diversity and quality of captions generated by a transformer-based vision-language model on remote sensing datasets?

B. Project Revision

Due to hardware limitations on training or fine-tuning a VLM, I had to change the project proposal to one that is based on inference and prompt engineering. The new research question is:

How does prompt structure affect the quality and reliability of image captions generated by a frozen PaliGemma model in remote sensing applications?

C. Objective and Significance

The primary objective of this revised project is to evaluate the impact of prompt phrasing on the captioning performance of a large vision-language model used in a zero-shot setting. Specifically, I aim to benchmark different prompt sets to generate 5 captions per image like in the dataset, and measure

their effect on output refusal rates, caption diversity, and relevance.

This is significant for two reasons: (1) It demonstrates how much can be gained through careful prompt design without any model fine-tuning, which is crucial in resource-limited environments. (2) It offers insights into using foundation models like PaliGemma for satellite imagery tasks, where pretraining data may not align perfectly with target domains.

D. Dataset and Preliminary Analysis

The RISC dataset consists of 44,521 satellite images, each paired with 5 human-written captions, totaling over 222,000 entries. Captions are short and often exhibit high overlap. For instance, many captions use repeated phrases such as "A building with a red roof." This redundancy limits model expressiveness. My initial analysis confirms an average caption length of 10 words and significant repetition across captions for a single image.

E. Methodology

This project uses the frozen `google/paligemma2-3b-mix-224` model for zero-shot image captioning via prompt engineering. I do not perform any fine-tuning. The model was run in my local environment on an NVIDIA RTX 2070 Super GPU. Caption generation was handled using HuggingFace Transformers' `generate()` method. All experiments used the same fixed subset of validation images from the dataset to ensure fair comparison across runs.

I designed three prompt sets (*basic*, *partial*, *descriptive*), each consisting of five distinct prompts. These were iteratively refined over three phases to reduce refusals and improve output relevance. The prompts used can be seen in Table I. Each image was processed with all prompt sets, generating 15 captions per image.

In addition, in an attempt to make the model generate meaningful and descriptive captions, a "word count prompt" was appended to the end of each of the final iteration prompts (e.g., "Use at least 10 words.").

Refusal responses (e.g., "Sorry, as a base VLM I am not trained...") were logged and used to quantify the impact of prompt wording on model behavior. Outputs were saved in structured JSON files, including prompts, generated captions, and reference captions for each image. This enables subsequent evaluation using standard caption quality metrics.

In Phase 3, I conducted a systematic ablation study on decoding parameters using the best-performing prompt set iteration. The goal was to understand how changes in *beam width* and *maximum output token length* affect both refusal behavior and caption quality.

Each generated caption was evaluated using industry-standard metrics:

- **BLEU** (Bilingual Evaluation Understudy): Measures n-gram precision between candidate and reference captions.

- **METEOR** (Metric for Evaluation of Translation with Explicit ORDERing): Considers both precision and recall, and includes stemming and synonym matching.
- **ROUGE-L** (Recall-Oriented Understudy for Gisting Evaluation): Evaluates the longest common subsequence between predicted and reference captions.

To ensure comparability, refusal responses were excluded from scoring. For each prompt set, aggregate scores and refusal rates were recorded across decoding strategies.

IV. RESULTS

A. Prompt Set Iterations

The initial evaluation focused on analyzing how different prompt sets and their refinements affected the image captions and model's willingness to produce responses. Refusal rates were used as the main metric, with a refusal defined as a generated caption consisting of fallback responses such as "Sorry, as a base VLM I am not trained to answer this question.", "unanswerable" or sometimes simply "no".

- **Iteration 1 (Initial Prompts):** 21.0% refusal rate (315/1500). Many prompts were vague or abstract, triggering generic denials.
- **Iteration 2 (Revised Prompts):** 28.9% refusal rate (434/1500). Some edits unintentionally resembled instruction-like phrasing.
- **Iteration 3 (Final Prompts):** Improving over the previous iteration as well as taking what worked better in the first iteration led to a **15.3%** refusal rate (229/1500), confirming that declarative and grounded phrasing improves success.
- **Iteration 3 + WCP:** Appending a word-count cue (e.g., "Use at least 10 words.") caused refusal rates to spike drastically—62.8% refusal rate (942/1500)—showing that the base PaliGemma model does not respond well to explicit instructions.

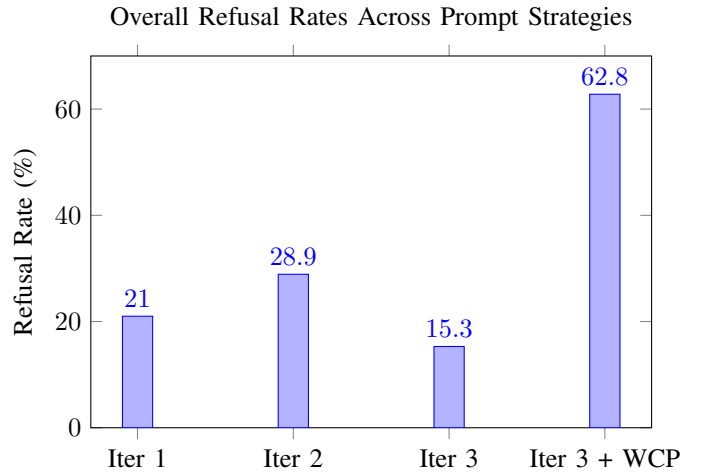


Fig. 1: Overall refusal rate per prompt iteration. Appending a word count instruction caused a large increase in refusals.

TABLE I: Prompt Variations Across Iterations

Prompt Set	Iteration 1	Iteration 2	Iteration 3
Basic	• Write a caption. • What do you see? • Summarize what’s visible. • Describe any visible activity. • Describe the visual content of this image.	• Write a caption. • Write a description. • Describe the image. • Caption the image. • Add a caption.	• Write a caption. • Write a description. • Describe the image. • Caption the image. • Add a caption.
Partial	• What subjects are visible? • Describe the type of area shown. • What is the background? • What are the contents of this image? • What are the features of this image?	• Write about the subjects in this image. • Describe the type of area shown. • Describe the background. • What are the contents of this image? • What are the features of this image?	• What subjects are visible? • Describe the type of area shown. • What is the background? • What is shown in this image? • What are the features of this image?
Descriptive	• What does this satellite image show? • Write a detailed description. • What is shown in this aerial view? • Describe all visible elements. • What is happening in this scene?	• Caption this satellite image. • Write a detailed description. • Describe the contents of this image in detail. • Write a detailed caption. • Caption what is happening in this scene?	• Caption this satellite image. • Write a detailed description. • Describe this image in detail. • Write a detailed caption. • Caption what is happening in this scene?

Each image was captioned 15 times (5 prompts $\times$ 3 sets) per iteration, and refusal counts were logged. Figure 1 presents the refusal rates per prompt set across four configurations: the three prompt iterations and an additional Word Count Prompt (WCP) condition appended to Iteration 3.

A fourth iteration was tested in Phase 3 by changing only the last descriptive prompt in Iteration 3 to “Describe this image.” but this resulted in a higher refusal rate of around 17%, and this iteration was discarded.

These results demonstrate that prompt engineering alone can meaningfully improve model reliability, and also highlight the limitations of instruction-free foundation models during inference.

B. Ablation Study: Decoding Parameters

In Phase 3, decoding configurations were tested to measure their influence on both refusal behavior and caption quality. Using the final prompt iteration, three decoding setups were evaluated:

- **Greedy decoding, max tokens 50** — baseline setting.
- **Beam search (beam size 5), max tokens 50** — to explore diversity and confidence weighting.
- **Greedy decoding, max tokens 100** — to test effect of longer generations.

Table II presents the average BLEU, METEOR, and ROUGE-L scores across all three prompt sets (excluding refusals), alongside refusal rates.

These results suggest that greedy decoding with a 50-token limit provides the best balance between refusal avoidance and caption quality. Beam search slightly improved METEOR scores but increased refusals, while increasing output length degraded ROUGE-L without reducing refusal rate.

TABLE II: Decoding Ablation: Average Scores Across All Prompt Sets

Config	BLEU	METEOR	ROUGE-L	Refusal Rate
Greedy, 50 tokens	0.0046	0.2636	0.1571	15.3%
Beam 5, 50 tokens	0.0043	0.2638	0.1473	20.6%
Greedy, 100 tokens	0.0049	0.2408	0.1344	15.3%

C. Prompt Set Comparison

To evaluate the impact of prompt phrasing independent of decoding strategy, I compared results across the three prompt sets—*basic*, *partial*, and *descriptive*—using the same final prompt iteration and decoding configuration (greedy, 50 tokens).

TABLE III: Prompt Set Performance Comparison (Final Config)

Prompt Set	BLEU	METEOR	ROUGE-L	Samples Used
Basic	0.0049	0.2806	0.1553	344
Partial	0.0017	0.2110	0.1626	109
Descriptive	0.0057	0.2904	0.1540	347

The *descriptive* set achieved the best BLEU and METEOR scores, indicating better fluency and semantic richness. The *partial* set had the highest ROUGE-L, likely due to tighter lexical overlap from structured phrasing, however the much lower samples used show that the set caused the most refusals. The *basic* set performed moderately across all metrics.

These results confirm that richer, more descriptive prompts elicit stronger captions, while narrowly scoped prompts yield more specific—but often shorter—outputs.

V. DISCUSSION

The results from the ablation study and prompt set comparison provide valuable insights into the behavior and limitations of the PaliGemma model in zero-shot captioning of satellite images. Two key aspects were studied: decoder parameter tuning and prompt strategy efficacy.

A. Decoder Parameters and Ablation

To satisfy the project requirement for exploring two hyperparameters or architectural changes, I conducted an ablation study by varying decoder settings during generation. Specifically, I experimented with `num_beams` (beam search width) and `max_new_tokens` (generation length cap). Higher beam values generally allow more exploration during decoding, while the token limit controls output length.

The results revealed marginal gains in evaluation metrics with more decoding flexibility. For instance, increasing `max_new_tokens` from 50 to 100 slightly improved BLEU scores, suggesting that longer generations contain more relevant content, but caused the other evaluation metrics to yield smaller scores. Increasing beam value from 1 to 5 increased latency and caused the number of refusals to increase. Thus, decoder tuning offers limited improvement and comes with trade-offs in runtime.

B. Prompt Set Comparison

The comparison across the three prompt sets (*basic*, *partial*, *descriptive*) highlighted significant differences in both refusal rates and caption quality. The **partial** set had the highest refusal rate, contributing only 109 usable generations—an indicator that its more analytical or open-ended questions confused the model. In contrast, the **basic** and **descriptive** sets were more successful in prompting relevant responses.

Evaluation via standard captioning metrics (BLEU, METEOR, ROUGE-L) showed that:

- **Descriptive** prompts produced the highest-quality captions overall, scoring best across most metrics.
- **Basic** prompts had a reasonable balance of low refusal rate and decent metric scores.
- **Partial** prompts had a better ROUGE-L score but high a refusal rate, making them least effective.

These findings suggest that descriptive prompts, when phrased declaratively and without instructional language, best align with the model’s learned priors, triggering more robust and complete image-grounded responses.

VI. CONCLUSION

This project investigated how prompt design affects the performance of a frozen transformer-based vision-language model (PaliGemma) in generating captions for satellite imagery. Due to hardware constraints, I shifted from fine-tuning to an inference-only approach focused on prompt engineering and decoding strategies.

Through multiple iterations of prompt sets, I observed that simple rewording of prompts—making them more grounded and less instructional—significantly reduced refusal rates and

improved caption success. Attempts to enforce longer captions via appended word count instructions led to a sharp increase in refusals, highlighting the model’s lack of instruction-following capability.

An ablation study on decoder parameters revealed that modest improvements in caption quality can be achieved by increasing token limits, while beam search depth had diminishing returns. However, most performance gains were driven by careful prompt design rather than generation tuning.

Ultimately, the best-performing configuration emerged from the *descriptive* prompt set paired with longer generation limits. This reinforces the importance of phrasing in eliciting meaningful zero-shot outputs from base vision-language models. Future work could explore combining prompt strategies with light-weight fine-tuning methods (e.g., adapters or LoRA) to bridge the gap between foundation model capabilities and task-specific needs.

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